

JAVA PRACTICAL RECORD

Additional Java Programs — Separate Experiments

EXPERIMENT 07 — CONDITIONAL STATEMENTS

Experiment No	07
Date	17/09/2026
Title	Conditional Statements

■ Objective / Aim

To write a Java program to demonstrate conditional statements using if-else and switch statements for even/odd checking, finding the largest number, assigning grades, and displaying days.

■ System Requirements

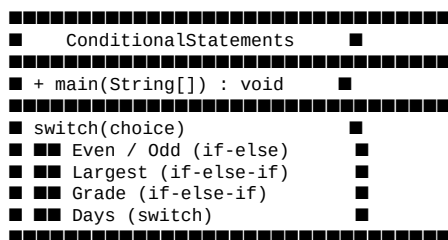
Hardware: Standard PC/Laptop, Core i3/i5 Processor, 4GB/8GB RAM.

Software: Windows/Linux/macOS, JDK 17/JDK 21, VS Code/Eclipse/IntelliJ IDEA/NetBeans.

■ Algorithm / Logic Description

1. Create a Scanner object to read input.
2. Display a menu containing four choices.
3. Use switch to select the required operation.
4. For choice 1, use if-else to check whether a number is even or odd.
5. For choice 2, use if-else-if to find the largest of three numbers.
6. For choice 3, use if-else-if to determine the grade from the percentage.
7. For choice 4, use switch to display the day for a number from 1 to 7.
8. Display the result and close the Scanner.

■ UML Class Diagram



■ Source Code

```
import java.util.Scanner;

public class ConditionalStatements {
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter your choice: 1.EvenOrOdd,2.Largest,3.Grade,4.Days");
        int ch=sc.nextInt();

        switch (ch)
        {
            case 1:

                System.out.println("Enter a number: ");
                int num=sc.nextInt();
                if(num % 2==0)
                {
                    System.out.println("Given number is a even number");
                }
                else
                {
                    System.out.println("Given number is a odd number");
                }
                sc.close();

                break;

            case 2:

                System.out.println("Enter three numbers : ");
                int a=sc.nextInt();
                int b=sc.nextInt();
```

```

int c=sc.nextInt();

if (a >= b && a >= c)
    System.out.println("largest: "+a);
else if (b >= a && b >= c)
    System.out.println("Largest is : "+b);
else
    System.out.println("Largest is : "+c);
break;

case 3:

    System.out.print("Enter Percentage: ");
    double per = sc.nextDouble();

    if (per >= 90)
        System.out.println("Grade A");
    else if (per >= 75)
        System.out.println("Grade B");
    else if (per >= 60)
        System.out.println("Grade C");
    else if (per >= 40)
        System.out.println("Grade D");
    else
        System.out.println("Fail");
    break;

case 4:

    System.out.print("Enter Day Number (1-7): ");
    int day = sc.nextInt();

    switch (day) {
        case 1:
            System.out.println("Monday");
            break;
        case 2:
            System.out.println("Tuesday");
            break;
        case 3:
            System.out.println("Wednesday");
            break;
        case 4:
            System.out.println("Thursday");
            break;
        case 5:
            System.out.println("Friday");
            break;
        case 6:
            System.out.println("Saturday");
            break;
        case 7:
            System.out.println("Sunday");
            break;
        default:
            System.out.println("Invalid Day");
    }
    break;

default:

    System.out.println("Enter a valid choice");
    break;
}
}
}

```

■ Compilation & Execution Commands

Save the file exactly as: ConditionalStatements.java
javac ConditionalStatements.java
java ConditionalStatements.java

■■ Output Screen

User may choose any menu option, so output depends on input.

■ Result

Thus, the Java program demonstrating conditional statements using if-else and switch was successfully implemented and executed.

EXPERIMENT 08 — LOOPING STATEMENTS

Experiment No	08
Date	17/09/2026
Title	Looping Statements

■ Objective / Aim

To write a Java program to demonstrate looping statements using for loops for multiplication tables, printing numbers, calculating the sum of n numbers, and generating Fibonacci series.

■ System Requirements

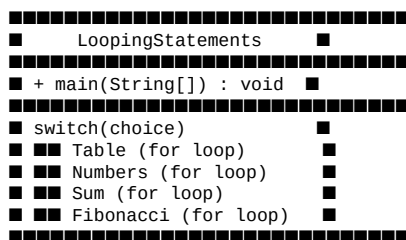
Hardware: Standard PC/Laptop, Core i3/i5 Processor, 4GB/8GB RAM.

Software: Windows/Linux/macOS, JDK 17/JDK 21, VS Code/Eclipse/IntelliJ IDEA/NetBeans.

■ Algorithm / Logic Description

1. Create a Scanner object.
2. Display a menu containing four choices.
3. Use switch to select an operation.
4. For choice 1, use a for loop to print the multiplication table from 1 to 10.
5. For choice 2, use a for loop to print numbers from 1 to N.
6. For choice 3, use a for loop to calculate the sum from 1 to N.
7. For choice 4, use a for loop to generate Fibonacci terms.
8. Display the result and close the Scanner.

■ UML Class Diagram



■ Source Code

```
import java.util.Scanner;

public class LoopingStatements {
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        System.out.println(" 1.Table\n 2.numbers (1-10)\n 3.sum of n numbers\n 4.fibonacci\n");
        System.out.println("Enter your choice : ");
        int ch=sc.nextInt();

        switch (ch)
        {
            case 1:

                System.out.println("Enter a number : ");
                int n=sc.nextInt();

                for(int i=1;i<=10;i++)
                {
                    System.out.println(n + "x" + i + "=" + (n * i));
                }
                break;
            case 2:

                System.out.print("Enter N: ");
                int num = sc.nextInt();

                for (int i = 1; i <= num; i++)
                    System.out.print(i + " ");
                break;
            case 3:
```

```

        System.out.print("Enter a number: ");
        int m = sc.nextInt();

        int sum = 0;

        for (int i = 1; i <= m; i++)
            sum += i;

        System.out.println("Sum = " + sum);
        break;
    case 4:

        System.out.print("Enter Number of Terms: ");
        int k= sc.nextInt();

        int first = 0, second = 1,next;

        for (int i = 1; i <= k; i++)
        {
            System.out.print(first + " ");
            next=first+second;
            first=second;
            second = next;
        }
        System.out.println();
        break;

    default:
        System.out.println("Enter a valid choice");
        break;
}
sc.close();
}
}

```

■ Compilation & Execution Commands

Save the file exactly as: LoopingStatements.java
javac LoopingStatements.java
java LoopingStatements.java

■■ Output Screen

User may choose any menu option, so output depends on input.

■ Result

Thus, the Java program demonstrating looping statements using for loops was successfully implemented and executed.

EXPERIMENT 09 — METHODS

Experiment No	09
Date	17/09/2026
Title	Methods

■ Objective / Aim

To write a Java program to demonstrate methods by implementing factorial, prime number checking, maximum of two numbers, and area of a circle.

■ System Requirements

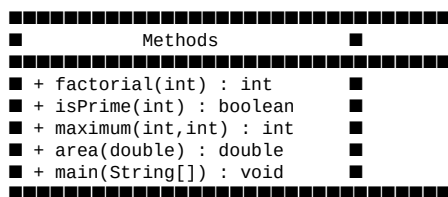
Hardware: Standard PC/Laptop, Core i3/i5 Processor, 4GB/8GB RAM.

Software: Windows/Linux/macOS, JDK 17/JDK 21, VS Code/Eclipse/IntelliJ IDEA/NetBeans.

■ Algorithm / Logic Description

1. Create a Scanner object.
2. Define a static method factorial() to calculate factorial.
3. Define a static method isPrime() to check whether a number is prime.
4. Define a static method maximum() to find the larger of two numbers.
5. Define a static method area() to calculate the area of a circle.
6. Display a menu and read the user's choice.
7. Call the appropriate method and display the result.
8. Close the Scanner.

■ UML Class Diagram



■ Source Code

```
import java.util.Scanner;

public class Methods {

    // Factorial
    static int factorial(int n) {
        int fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }

    // Prime
    static boolean isPrime(int n) {
        if (n <= 1)
            return false;

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0)
                return false;
        }
        return true;
    }

    // Maximum
    static int maximum(int a, int b) {
        return (a > b) ? a : b;
    }

    // Area of Circle
    static double area(double r) {
        return 3.14 * r * r;
    }
}
```

```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.println("1. Factorial");
    System.out.println("2. Prime Check");
    System.out.println("3. Maximum of Two Numbers");
    System.out.println("4. Area of Circle");
    System.out.print("Enter your choice: ");

    int ch = sc.nextInt();

    switch (ch)
    {

        case 1:
            System.out.print("Enter a number: ");
            int n = sc.nextInt();
            System.out.println("Factorial = " + factorial(n));
            break;

        case 2:
            System.out.print("Enter a number: ");
            int p = sc.nextInt();
            if (isPrime(p))
                System.out.println("Prime Number");
            else
                System.out.println("Not a Prime Number");
            break;

        case 3:
            System.out.print("Enter first number: ");
            int a = sc.nextInt();
            System.out.print("Enter second number: ");
            int b = sc.nextInt();
            System.out.println("Maximum = " + maximum(a, b));
            break;

        case 4:
            System.out.print("Enter radius: ");
            double r = sc.nextDouble();
            System.out.println("Area = " + area(r));
            break;

        default:
            System.out.println("Invalid Choice");
    }

    sc.close();
}
}

```

■ Compilation & Execution Commands

Save the file exactly as: Methods.java
javac Methods.java
java Methods.java

■■ Output Screen

User may choose any menu option, so output depends on input.

■ Result

Thus, the Java program demonstrating methods was successfully implemented and executed.

EXPERIMENT 10 — STUDENT INFORMATION

Experiment No	10
Date	17/09/2026
Title	Student Information

■ Objective / Aim

To write a Java program to read student information, calculate the total marks and percentage, and display the student details.

■ System Requirements

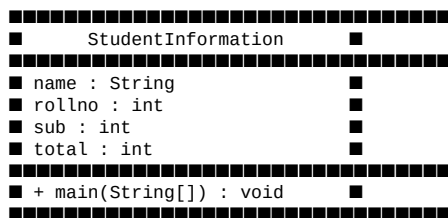
Hardware: Standard PC/Laptop, Core i3/i5 Processor, 4GB/8GB RAM.

Software: Windows/Linux/macOS, JDK 17/JDK 21, VS Code/Eclipse/IntelliJ IDEA/NetBeans.

■ Algorithm / Logic Description

1. Create a Scanner object.
2. Read the student's name.
3. Read the roll number.
4. Read the number of subjects.
5. Initialize total marks to zero.
6. Use a for loop to read marks for each subject and add them to total.
7. Calculate percentage as total marks divided by the number of subjects.
8. Display the student's name, roll number, total marks, and percentage.
9. Close the Scanner.

■ UML Class Diagram



■ Source Code

```
import java.util.Scanner;

public class StudentInformation
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the name: ");
        String name=sc.nextLine();
        System.out.println("Enter the roll number: ");
        int rollno=sc.nextInt();
        System.out.println("Enter the no of subjects: ");
        int sub=sc.nextInt();
        int total=0;
        for(int i=1;i<=sub;i++)
        {
            System.out.println("Enter the marks of subject "+i+":");
            int marks=sc.nextInt();
            total=total +marks;
        }
        int percentage;
        percentage=(total)/sub;
        System.out.println("The student name is : "+name);
        System.out.println("Student roll no is : "+rollno);
        System.out.println("Total marks is: "+total);
        System.out.println("The percentage of the student marks is: "+percentage);
        sc.close();
    }
}
```

■ Compilation & Execution Commands

Save the file exactly as: StudentInformation.java
javac StudentInformation.java
java StudentInformation.java

■■ Output Screen

Example input:

Sai
18
3
80
90
70

Example output:

The student name is : Sai
Student rool no is : 18
Total marks is: 240
The percentage of the student marks is: 80

■ Result

Thus, the Java program to read and display student information and calculate total marks and percentage was successfully implemented and executed.